

Streptococcus A Lab-On-Time™ Test Device (BMT 51084)

CLINICAL VALIDATION REPORT

1. OBJECTIVE

The purpose of the test is to validate the performance of BMT's Streptococcus A Lab-On-Time™ Test Device and evaluate the usability of this rapid test in the hands of professional users and their ability to properly perform it based on the provided IFU and to correctly interpret its result.

2. OVERVIEW

Streptococcus pyogenes (Strep A) is non-motile gram-positive cocci, which contains the Lancefield group A antigens that can cause serious infections such as pharyngitis, respiratory infection, impetigo, endocarditis, meningitis, puerperal sepsis, and arthritis.

Pharyngitis (strep throat) is a common infectious disease among 5–15-year-old and characterized by the rapid appearance of symptoms including: headache, fever, throat and swallowing pain and stomachache. Throat inspection usually reveals erythema at the back of the throat and the larynx with white effluent, swollen tonsils and sometimes also painful and enlarged neck lymph nodes. Left untreated, strep A infections may lead to serious complications, including rheumatic fever and peritonsillar abscess. The traditional identification procedures for Group A Streptococci infection involve the isolation and identification of viable organisms using techniques that require 24 to 48 hours or longer.

The Strep Lab-on-Time™ Strep A Rapid Test is an integrated device for throat swab sample collection and analysis, employing a qualitative lateral flow immunoassay for the rapid (5-10min) detection of Strep A carbohydrate antigen. In this test, antibody specific to Strep A carbohydrate antigen is coated on the test line region of the test. During testing, the extracted throat swab specimen reacts with an antibody to Strep A that is coated onto gold nanoparticles. The mixture migrates by capillary forces along the membrane to react with the antibody to Strep A on the membrane and generate a colored line in the test line region. The presence of this colored line in the test line region indicates a positive result, while its absence indicates a negative result. To serve as a procedural control, a colored line will always appear in the control line region, indicating that proper volume of specimen has been added and membrane wicking has occurred. This test is qualitative, displaying an analytical sensitivity (limit of detection) of 1×10^5 Strep A organisms. It is not intended for the quantitative determination of Strep A bacteria.

Use of this device in the hospital, clinic, or even self-use by the patient will provide a rapid result indicating the need for antibiotic treatment (such as by Penicillin and its derivatives) and may eliminate the need for complicated, time-consuming, and expensive tests, such as PCR or microbiological culture.

3. SPECIFIC STUDY GOALS

- Directly compare between the results obtained using the Strep A Lab-On-Time™ Rapid Test Device with freshly obtained throat swab samples from symptomatic patients and results obtained with

matching samples using a lab-based culture procedure.

- Determine test sensitivity and specificity in comparison with the results obtained by the microbiological laboratory.
- Evaluate the usability of BMT's Strep A Test Device for the purpose of recommending this device for self-testing in the hands of lay untrained users.

4. VALIDATION STUDY OUTLINE

The test was validated at the Tzafon Medical Center, Poriya, Israel in Feb-Mar 2023 under the supervision of Prof. Avi Peretz (PI) using kits provided by BMT Diagnostics (BMT 51084, LOT , Expiry).

Study recruitment was performed using a Consent Form that was signed by the patients. All data regarding the patients was encoded and available only to the PI. Age of recruited patents was over 3 years, and the study was open to both women and man. Excluded from this study were individuals who lack judgement or are employees/students supervised by the PI. Also excluded are patients who have received antibiotics of any kind 24 (or less) prior to test performance.

The study was conducted on 181 patients, of which 70 were positive and 111 negatives to Strep A according to the microbiological culture test. The test was performed according to the test manufacturer's protocol (IFU). The comparison was made using duplicate patient samples sent to the microbiological lab for culture and analysis.

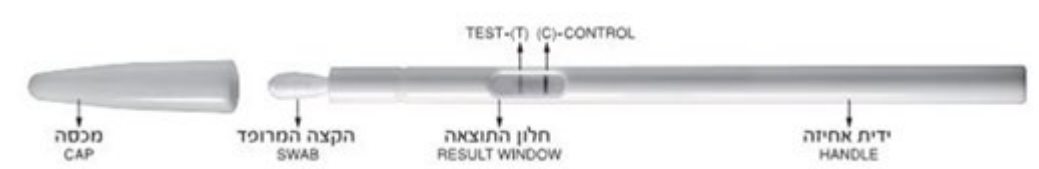
The test was performed by a lab technician, simulating self-testing conditions by collecting a throat sample from the patient by the device swab and following the protocol provided in the IFU. A matching throat swab specimen was collected and used for microbiological validation of the results.

The lab technician didn't have any previous experience with performing this test and was therefore only exposed to and educated by the IFU. This assured that the technician operated the test as a lay person would, thereby simulating self-testing.

The following information was collected for each recruited patient: test results (irrespective of the current study), age and sex, and how long prior to the test has the patient observed fever and throat pain. Test validation results will be reported to the Israeli Ministry of Health according to the clinical validation protocol, including the observed test sensitivity and specificity.

5. TEST PERFORMANCE

The following picture details the Strep A Lab-On-Time Device, showing the integrated swab and results window:



Strep A Lab-On-Time Device Components

The Strep A Lab-On-Time device is indicated for professional use by a trained medical practitioner (nurse/physician) in the physician's office or in the lab. The test is performed by extracting the device and reagent tube from their respective aluminum pouches, followed by removing the device cap, collecting the throat sample using the swab end of the device and transferring the device into the reagent tube such that the swab end of the device is being soaked in the tube reagent. The device end is then mixed in the tube for around 10 seconds, followed by reading the test results in the device's results window (see also product IFU in the Appendix for additional detail).

6. RESULTS

Table1 – Lab-On-Time Strep A Rapid Test Results compared with microbiological culture:

Method		Culture			Total Results
Lab-On-Time Strep A rapid test device	Results	Positive	Few colonies	Negative	
	Positive	62	0	0	62
	Negative	2	6	111	119
Total results		64	6	111	181

As shown in the above table, some of the microbiological cultures for Strep A positive patients displayed only a few (less than 10) bacterial colonies, indicating that the initial bacterial load was low.

Table2 – Summary of Lab-On-Time Strep A Test Results:

	Overall	Lower C.I. (95%)	Upper C.I. (95%)	Excluding Rare Culture	Lower C.I. (95%)	Upper C.I. (95%)
Relative Sensitivity	88.6%	81.1%	96.0%	96.9%	92.6%	100.0%
Relative Specificity	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Relative Accuracy	95.6%			98.9%		
PPV	1.00			1.00		
NPV	0.93			0.98		

7. OBSERVATIONS AND CONCLUSION

The above evaluation study compared the diagnostic test results obtained using BMT Diagnostics Lab-

On-Time Strep A Rapid Test Device with culture test results obtained by the microbiological laboratory using standard culture techniques performed using 181 throat swab samples obtained from healthy (111) and strep A patients. From this study, it is concluded that the specificity of BMT's rapid test is 100%, whereas the overall sensitivity was found to be 88.6%.

Further analysis performed according to the number of strep A colonies appearing following microbiological culture demonstrated that when considering only colony-rich bacterial culture, the sensitivity of BMT's rapid test increased to 96.9%.

BMT's rapid test device has been shown to be very user friendly. Test performance is very intuitive and involves only a single step of sample collection and placement of the test device in the provided, pre-filled reagent tube. Test results were clear and obtained in less than 10 min following throat sample collection by the device's swab.

It is my professional opinion that BMT Diagnostics Lab-On-Time Strep A Rapid Test Device is very similar in its results to those obtained by bacterial culture and could be easily adopted by professional users due to its simplicity, ease of use, and clear results reading.

Signature:



Prof. Avi Peretz

Director Research Institute and Clinical Microbiology Clinic

Tzafon Medical Center, Poriya, Israel

Appendix – Results

Table 1 – Culture-positive subjects

Sample ID	M/F	Antigen Test Result	Culture result	Comments
#00 1	F	Positive	Positive	
#00 2	M	Positive	Positive	
#00 3	F	Negative	Positive	Poor growth
#00 9	F	Positive	Positive	
#00 15	M	Positive	Positive	
#00 16	M	Positive	Positive	
#00 24	M	Positive	Positive	
#00 25	F	Positive	Positive	
#00 32	M	Positive	Positive	
#00 33	F	Positive	Positive	
#00 34	F	Positive	Positive	
#00 35	M	Positive	Positive	
#00 36	M	Positive	Positive	
#00 37	F	Positive	Positive	
#00 38	M	Positive	Positive	
#00 39	F	Positive	Positive	
#00 40	F	Positive	Positive	
#00 41	M	Positive	Positive	
#00 42	M	Positive	Positive	
#00 43	F	Positive	Positive	
#00 44	M	Positive	Positive	

#00 45	M	Positive	Positive	
#00 46	F	Positive	Positive	
#00 47	F	Positive	Positive	
#00 49	M	Negative	Positive	Poor growth
#00 55	M	Positive	Positive	
#00 61	F	Positive	Positive	
#00 65	F	Positive	Positive	
#00 70	F	Positive	Positive	
#00 71	M	Positive	Positive	
#00 77	F	Positive	Positive	
#00 82	F	Positive	Positive	
#00 85	F	Positive	Positive	
#00 86	F	Positive	Positive	
#00 87	F	Positive	Positive	
#00 88	F	Positive	Positive	
#00 89	M	Positive	Positive	
#00 90	M	Negative	Positive	Poor growth
#00 93	M	Negative	Positive	Poor growth
#00 94	F	Positive	Positive	
#00 98	M	Positive	Positive	
#00 99	F	Positive	Positive	
#00 100	M	Positive	Positive	
#00 101	M	Positive	Positive	
#00 103	M	Positive	Positive	
#00 107	F	Negative	Positive	Rich growth, increased gag reflex observed
#00 112	M	Positive	Positive	

#00 115	M	Negative	Positive	Poor growth
#00 123	M	Positive	Positive	
#00 127	M	Positive	Positive	
#00 132	M	Positive	Positive	
#00 136	F	Negative	Positive	Rich growth
#00 139	M	Positive	Positive	
#00 142	F	Positive	Positive	
#00 143	F	Positive	Positive	
#00 146	M	Positive	Positive	
#00 147	M	Negative	Positive	Poor growth, possible antibiotic treatment
#00 151	F	Positive	Positive	
#00 155	M	Positive	Positive	
#00 156	F	Positive	Positive	
#00 157	F	Positive	Positive	
#00 158	F	Positive	Positive	
#00 159	F	Positive	Positive	
#00 161	M	Positive	Positive	
#00 165	M	Positive	Positive	
#00 169	M	Positive	Positive	
#00 173	M	Positive	Positive	
#00 175	F	Positive	Positive	
#00 176	F	Positive	Positive	
#00 179	F	Positive	Positive	

Table 2 – Culture-negative subjects

Sample ID	M/F	Antigen Test Result	Culture result	Comments
#00 4	M	Negative	Negative	
#00 5	F	Negative	Negative	
#00 6	M	Negative	Negative	
#00 7	M	Negative	Negative	
#00 8	M	Negative	Negative	
#00 10	M	Negative	Negative	
#00 11	F	Negative	Negative	
#00 12	F	Negative	Negative	
#00 13	F	Negative	Negative	
#00 14	M	Negative	Negative	
#00 17	M	Negative	Negative	
#00 18	M	Negative	Negative	
#00 19	M	Negative	Negative	
#00 20	M	Negative	Negative	
#00 21	M	Negative	Negative	
#00 22	F	Negative	Negative	
#00 23	F	Negative	Negative	
#00 26	M	Negative	Negative	
#00 27	F	Negative	Negative	
#00 28	F	Negative	Negative	
#00 29	M	Negative	Negative	
#00 30	F	Negative	Negative	
#00 31	M	Negative	Negative	

#00 48	M	Negative	Negative	
#00 50	M	Negative	Negative	
#00 51	M	Negative	Negative	
#00 52	M	Negative	Negative	
#00 53	M	Negative	Negative	
#00 54	M	Negative	Negative	
#00 56	M	Negative	Negative	
#00 57	M	Negative	Negative	
#00 58	M	Negative	Negative	
#00 59	M	Negative	Negative	
#00 60	M	Negative	Negative	
#00 62	F	Negative	Negative	
#00 63	F	Negative	Negative	
#00 64	F	Negative	Negative	
#00 66	M	Negative	Negative	
#00 67	F	Negative	Negative	
#00 68	F	Negative	Negative	
#00 69	F	Negative	Negative	
#00 72	M	Negative	Negative	
#00 73	F	Negative	Negative	
#00 74	F	Negative	Negative	
#00 75	M	Negative	Negative	
#00 76	F	Negative	Negative	
#00 78	F	Negative	Negative	
#00 79	F	Negative	Negative	
#00 80	M	Negative	Negative	

#00 81	M	Negative	Negative	
#00 83	F	Negative	Negative	
#00 84	F	Negative	Negative	
#00 91	M	Negative	Negative	
#00 92	M	Negative	Negative	
#00 95	F	Negative	Negative	
#00 96	M	Negative	Negative	
#00 97	F	Negative	Negative	
#00 102	F	Negative	Negative	
#00 104	M	Negative	Negative	
#00 105	F	Negative	Negative	
#00 106	F	Negative	Negative	
#00 108	F	Negative	Negative	
#00 109	M	Negative	Negative	
#00 110	M	Negative	Negative	
#00 111	M	Negative	Negative	
#00 113	F	Negative	Negative	
#00 114	F	Negative	Negative	
#00 116	F	Negative	Negative	
#00 117	M	Negative	Negative	
#00 118	M	Negative	Negative	
#00 119	M	Negative	Negative	
#00 120	F	Negative	Negative	
#00 121	F	Negative	Negative	
#00 122	F	Negative	Negative	
#00 124	M	Negative	Negative	

#00 125	F	Negative	Negative	
#00 126	M	Negative	Negative	
#00 128	M	Negative	Negative	
#00 129	M	Negative	Negative	
#00 130	M	Negative	Negative	
#00 131	F	Negative	Negative	
#00 133	M	Negative	Negative	
#00 134	M	Negative	Negative	
#00 135	F	Negative	Negative	
#00 137	M	Negative	Negative	
#00 138	M	Negative	Negative	
#00 140	M	Negative	Negative	
#00 141	M	Negative	Negative	
#00 144	F	Negative	Negative	
#00 145	F	Negative	Negative	
#00 148	M	Negative	Negative	
#00 149	F	Negative	Negative	
#00 150	F	Negative	Negative	
#00 152	F	Negative	Negative	
#00 153	F	Negative	Negative	
#00 154	F	Negative	Negative	
#00 160	F	Negative	Negative	
#00 162	F	Negative	Negative	
#00 163	F	Negative	Negative	
#00 164	F	Negative	Negative	
#00 166	M	Negative	Negative	

#00 167	F	Negative	Negative	
#00 168	F	Negative	Negative	
#00 170	M	Negative	Negative	
#00 171	F	Negative	Negative	
#00 172	F	Negative	Negative	
#00 174	F	Negative	Negative	
#00 177	F	Negative	Negative	
#00 178	M	Negative	Negative	
#00 180	M	Negative	Negative	
#00 181	F	Negative	Negative	



מעבדה למיקרוביולוגיה

מנהל פרופ' אבי פרץ

Table 3 – Instructions For Use (IFU)